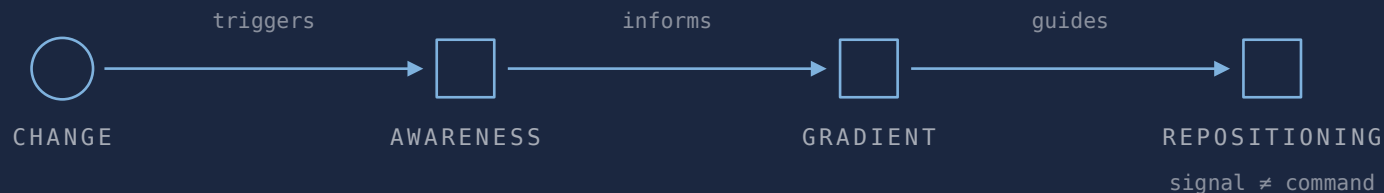

Ecosystem Positioning for Agentic Systems

A proposed architecture for participants that must stay situated, and requalify, as authority, evidence, roles and the ecosystem itself change.



WORKING ARCHITECTURE

Proposal
Open for review.

CREATED BY

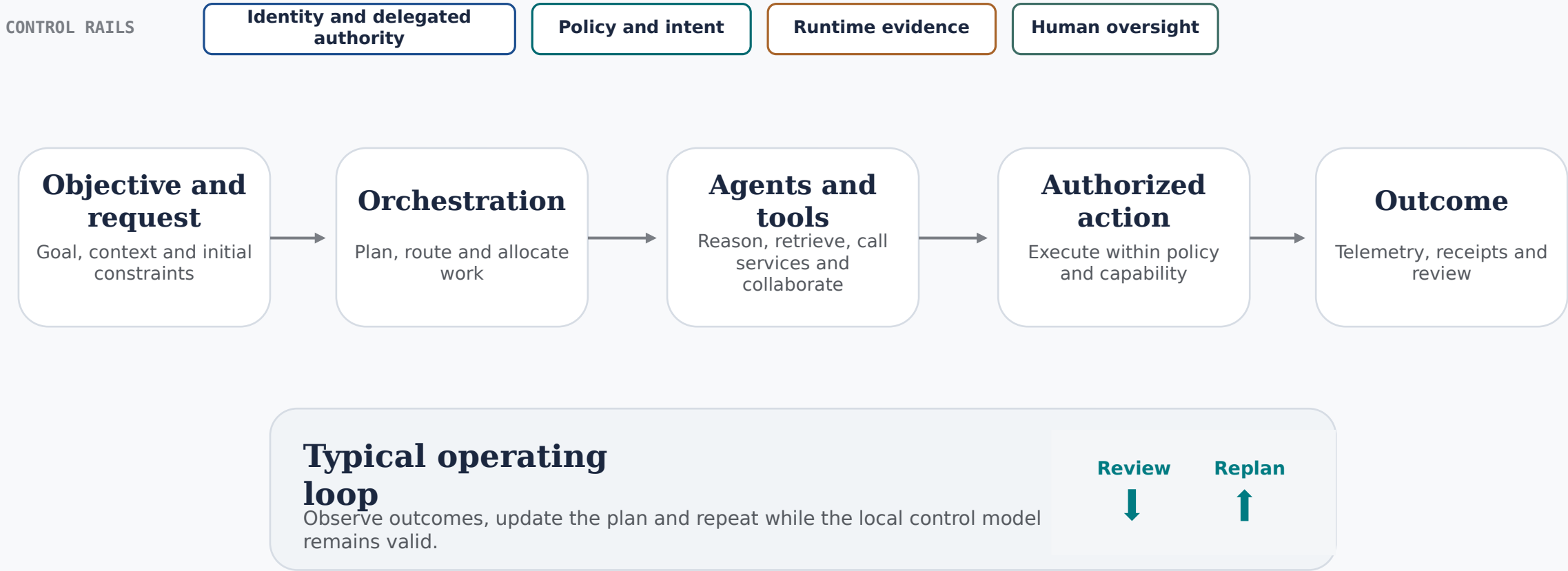
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CREATIVE COMMONS · ATTRIBUTION

V1.0 · 2026-09-23 · CANONICAL WORKING PROPOSAL

Current Reference Architecture for Agentic AI

A composite of current practice: the strong baseline EA is benchmarked against (arms B1/B2), not a strawman.



The model works while roles, boundaries and the surrounding ecosystem remain legible to the controls that govern the workflow.

Benchmark, differential and public work

Existing mechanisms remain primary for their respective primitives; the proposal addresses the composition and requalification gap between them.

BENCHMARK AGAINST THE STRONG BASELINE

IDENTITY · AUTHORITY · ATTESTATION

Existing: principal binding, mandate, authorization, integrity and evidence appraisal.

EA / Positioning: preserves scope and residual; local validity is not silently promoted to composed-decision validity.

A2A · MCP · AGENT FRAMEWORKS

Existing: discovery, transport, tools, context, orchestration, messages and artifacts.

EA / Positioning: qualifies what a received result establishes and carries provenance, UNKNOWNs and revalidation conditions.

UQ · CONFIDENCE · EVIDENCE FUSION

Existing: calibration, confidence signalling and inherited uncertainty propagation.

EA / Positioning: adds A/B/C/D, window boundaries, structural residual and non-fungible composition across decision domains.

SIGNALLING · INCIDENT DEFENSE

Existing: warning exchange, corroboration, blast radius, containment and recovery.

EA / Positioning: also addresses the non-incident case where honest, locally correct components remain systemically insufficient.

DIFFERENTIAL UNDER TEST Scope-indexed, non-fungible epistemic composition with bounded requalification across independently governed participants.

PUBLIC PRE-STANDARDIZATION FOOTPRINT

FG-TIDA · public contributor-level work

#13 · Ecosystem / signal interface

Systemic-capacity / handoff architecture and interoperability with signalling / defense.

#16 · Operational human oversight

EA assessment feeds the intervention path; human-capacity state feeds ecosystem assessment.

#6 / #21 · verdict + evidence interfaces

Bounded verdict, scope and population-level evidence can remain qualified across handoffs.

External industry case mapping · Panodyyssey

Authority, regime/context and evidence-to-decision challenges mapped against an external industry case; separate from FG-TIDA adoption.

PUBLIC / CANONICAL ROUTES

[Structural Awareness](#) · [Ecosystem Positioning](#)
[Ecosystem Awareness](#) · [Regime Awareness](#) · [MSCA Benchmark v0.2](#) · [FG-TIDA application package](#)

CLAIM BOUNDARY

Working architecture / public pre-standardization. No ITU-T adoption or endorsement is claimed.

Structural problems and current limits

Direct rules and local checks lose explanatory power when participants, roles and dependencies change faster than the control model.

Agent multiplication

More agents, tools and temporary delegates create interaction paths that no operator can review one by one.

Role and identity drift

One participant can hold hybrid roles, change memberships or act under a different authority across interactions.

Ecosystem change

Organizations, dependencies, service availability and external conditions evolve while the system is operating.

Indeterminacy

A local result may be valid but insufficient for the wider decision. Missing scope, provenance or dependency state must remain explicit.

Interpretation gap

Identity, policy and attestation can remain technically valid while their practical meaning changes in the active ecosystem.

Limits of AI oversight

Automated review often inherits the same observation window. Human review adds authority and judgment, but attention, time and context remain finite.

EXISTING PRIMITIVES SPIFFE · identity/authentication | RFC 8693 · subject/actor token exchange | RATS · attestation/appraisal | NIST 2026 · active agent-IAM work
REMAINING GAP **Preserve what remains valid, knowable and actionable as authority, evidence and context change across handoffs.**

STRUCTURAL CONSEQUENCE

Local correctness and local compliance do not by themselves establish that the composed system is situated correctly for the current decision.

Failure progression: mechanisms and failure types

The ecosystem can drift while local controls keep returning valid-looking answers. The key question is whether uncertainty remains visible or is converted into unsupported closure.

INITIAL STATE

Legible roles and boundaries

Identity and authority are current
Policies match the operating context
Oversight can follow the material paths

DRIFT

The ecosystem changes

Agents multiply and change roles
Boundaries and memberships shift
Signals inherit hidden dependencies
Rules and reviews lag behind

FAILURE

Local closure, systemic indeterminacy

A local PASS travels beyond its scope
Uncertainty and lineage disappear
AI review can repeat the same frame
Human attention saturates

CONDITION AND FAILURE TYPES

TYPE 0 · Structural non-determination

Residual indeterminacy is not removable inside the current representation / capability boundary. More processing of the same evidence cannot manufacture the missing distinction.

Response: preserve the residual, narrow the claim, add a genuinely new information channel or redesign the boundary.

TYPE 1 · Unbounded unresolved uncertainty

Uncertainty is recognized, but no legitimate bounded closure is reached: the system keeps waiting, searching or escalating without an escape condition (indefinite HOLD, paralysis). HOLD is not a safe terminal state.

Response: establish bounded closure: a stop condition plus a qualified posture that keeps the residual explicit.

TYPE 2 · False certainty

The system closes by discarding what it does not know: missing input treated as present, low confidence promoted to certainty, a local result treated as globally valid (forced PASS/FAIL, false closure).

Response: restore uncertainty, scope and provenance, then re-enter the affected scope.

Illustrative failure scenarios

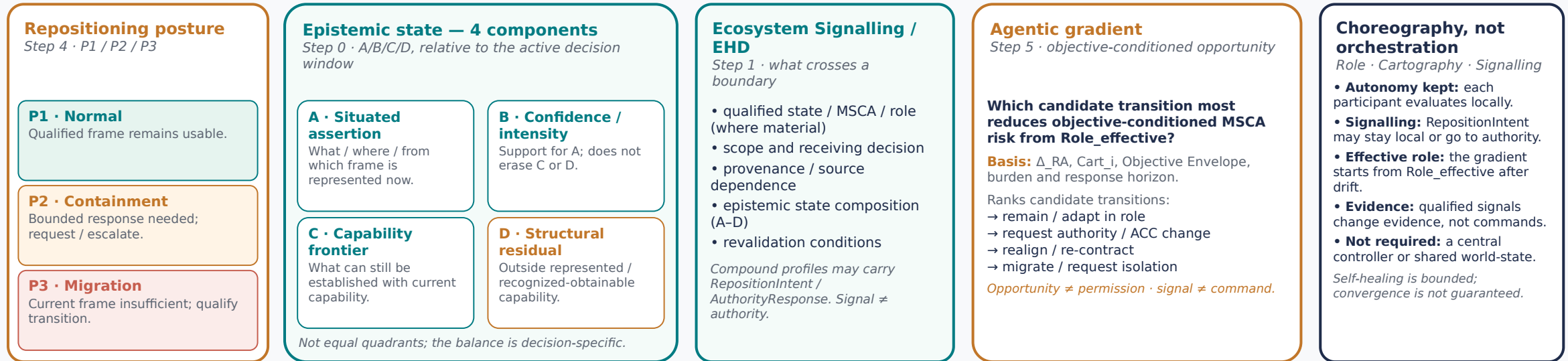
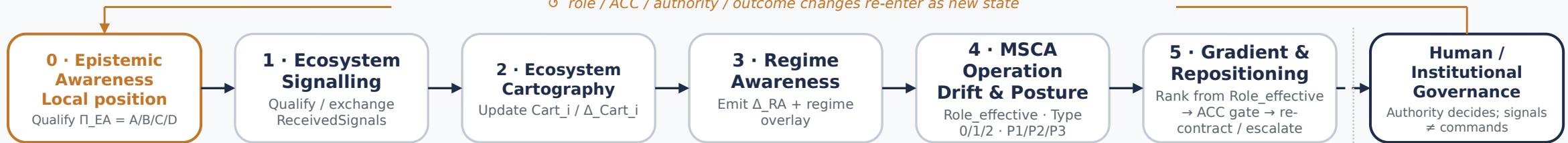
100 Million Tokens — Enterprise evidence compression and false closure under scale. [GitHub reference](#)

Smart-City Mobility Chaos — Distributed mobility drift and cross-actor systemic divergence. [GitHub reference](#)

From unresolved ecosystem change to qualified repositioning

Working proposal, not an adopted standard. Top row: the process sequence. Panels below: expansion of the numbered steps.

↻ role / ACC / authority / outcome changes re-enter as new state

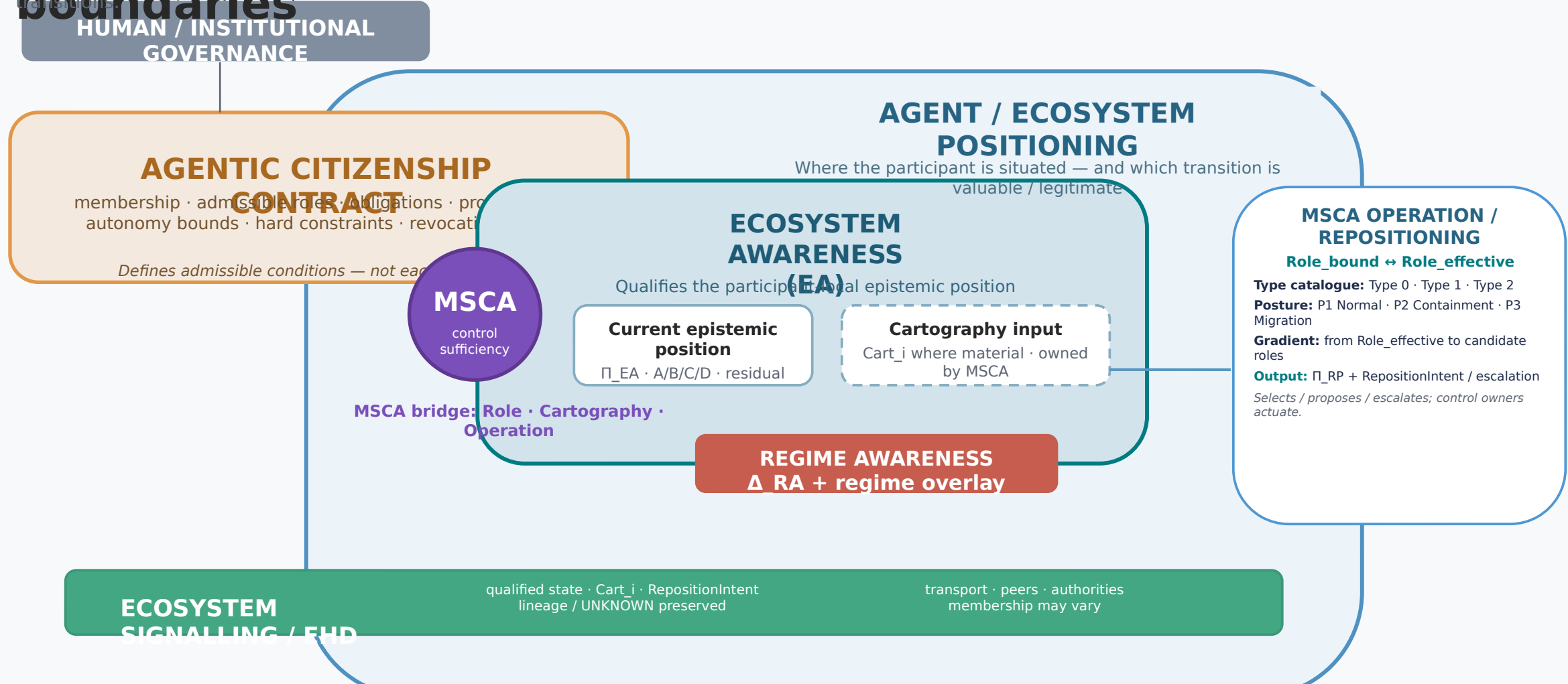


MSCA · Minimum Sufficient Control Architecture

MSCA static + dynamic set — S/E/C/P/M kernel · Architectural Role · Ecosystem Cartography · Operation/Repositioning
 Re-assessed when frame, role, envelope or authority changes; MSCA never creates authority.

Architecture map — where each mechanism lives - boundaries

Citizenship constrains admissibility; Role fixes bound function; EA qualifies knowledge; Cartography maps dependencies; Repositioning proposes / escalates transitions.



Key boundary: Citizenship constrains admissibility. Role binds function. EA qualifies knowledge. Cartography maps dependencies. Repositioning selects / proposes / escalates; control owners act. Signalling crosses boundaries.

Ecosystem Awareness — subsystem inputs and outputs

EA interface slice. Green = inputs to EA; orange = EA outputs / requests — never commands or authority.

Consumes existing identity, authorization and attestation mechanisms as inputs; does not replace them.

Human / Institutional Governance

Objectives, criticality, tolerated residual, deadlines and control options.

04 · 01 Mission / Objective · IF-S13 Trust framework

IN → to EA

objective and material decision domains · criticality, deadline, tolerated residual · authority refs

OUT ← from EA

qualified affected state · requalification / escalation request

External Agents / A2A / Epistemic Signaling

04 · 02 · 06 · IF-S11 Ecosystem signalling

IN ↓ to EA

EHD / envelope · source and provenance · freshness · corroboration history

OUT ↑ from EA

External Epistemic Envelope (F8) · targeted corroboration · source-dependence mark

IN ← to EA

Attestation Result · freshness / replay · appraisal relationship · limitations

OUT → from EA

re-attest or independently rooted attestation · reduce reliance to attested scope

Attestation / Runtime Evidence

Attestation results and verifier-side appraisal of runtime and interaction state.

04 · IF-S3 Attestation · IF-S5 Evidence appraisal

Identity & Delegated Authority

Who/what is acting, on whose behalf, under which mandate and scope.

04 · IF-S1 Identity · IF-S2 Authority / Delegation

IN → to EA

principal binding · grant scope and limits · validity / revocation · contested or absent state

OUT ← from EA

authority / binding UNKNOWN where binding · narrow to qualified mandate · refresh

ECOSYSTEM AWARENESS

participant-local qualification layer

$\Pi_{EA} = [A, B, C, D]$

W(d,t) · provenance · residual

ReceivedSignals qualification

Cart_i / Δ_{RA} inputs where material requalification / evidence requests

→ **MSCA Operation / Repositioning**

IN ← to EA

response capability and reach · actionability window · residual exposure · rollback state

OUT → from EA

affected scope · requalification / containment request

Ecosystem Defense / Self-Healing / Recovery

Containment, revocation, recovery and migration under legitimate authority.

04 · IF-S12 Enforcement / Containment / Migration

Policy / Intent / Constraints

Versioned policy and intent references and their runtime conformance.

04 · IF-S4 Policy / Intent conformance

IN → to EA

versioned policy ref · verdict permit ... indeterminate · scope examined · competing directives

OUT ← from EA

preserve INDETERMINATE · verdict scope insufficient · re-evaluate / independent evaluator

IN ↑ to EA

result and execution status · drift / telemetry · verifiable action record

OUT ↓ from EA

outcome reconciliation · qualified effect state · F8 statement for record

IN ← to EA

human-capacity state · reviewer role and authority · response window · intervention outcome

OUT → from EA

targeted review for the affected domain · capacity-insufficient flag · Type 1 stop boundary

Human Oversight / Assurance

Review, intervention and authority beyond autonomous limits.

04 · IF-S6 Human oversight capacity

Execution / Outcome / Audit

Tools, outcomes, telemetry and action records.

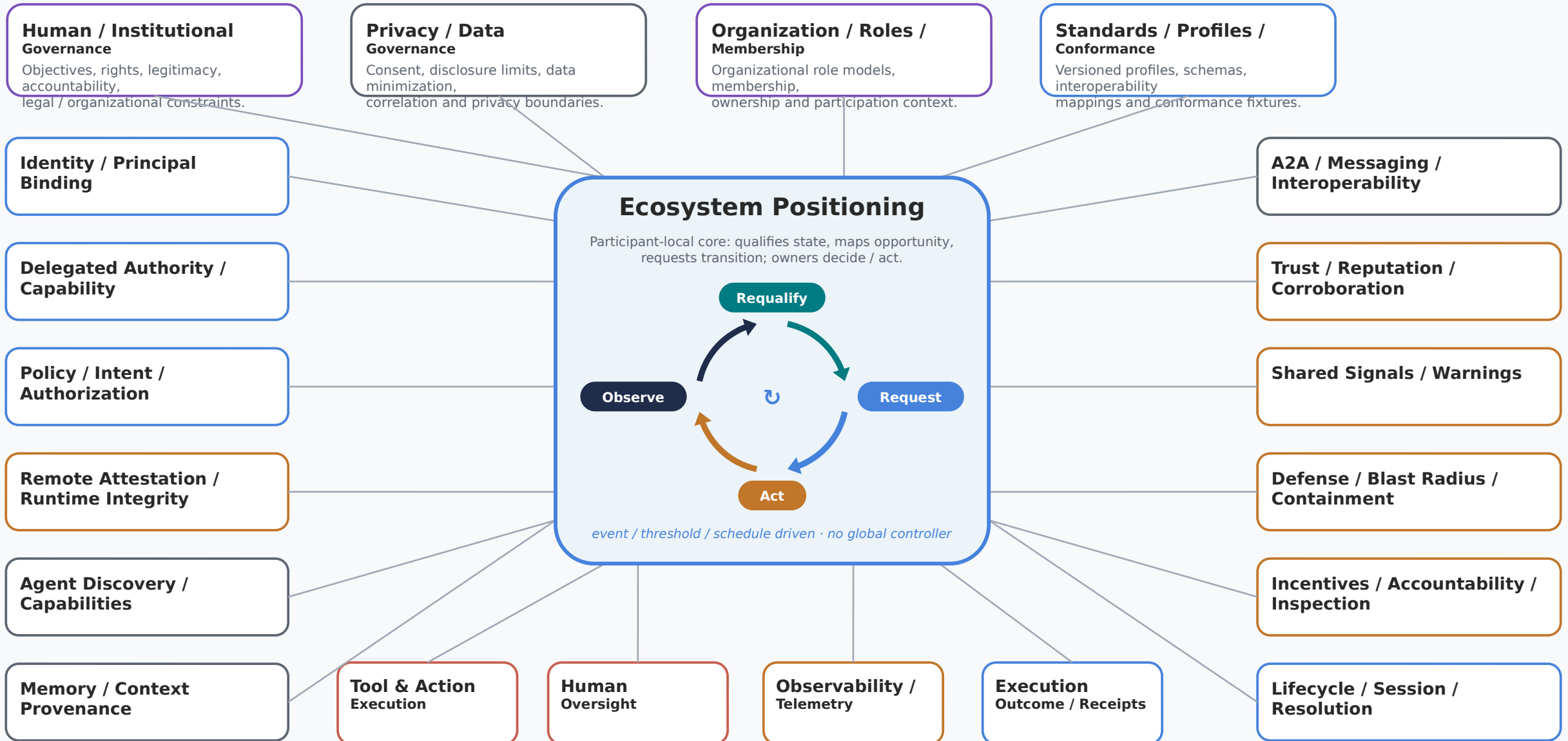
04 · 03 · 05 · IF-S7 · IF-S8

IN · to EA inputs listed in doc 04 for that interface **OUT · from EA** EA outputs to that function **04 · codes** interface IDs 01-06 / IF-S1-S13

Not shown: 04 context / memory / retrieval, IF-S9 evaluation, IF-S10 privacy. Final P1/P2/P3 closure belongs to MSCA Operation/Repositioning.

Ecosystem Positioning Architecture — Extended Situational Core

Agentic-AI, decentralized-identity (DIF) and ecosystem-defense components around one core: Ecosystem Positioning.



Reading rule: outer boxes are familiar agentic-AI capabilities. Positioning composes their decision-relevant state into a participant-local frame and requests requalification; owners decide.